

Jiwon Ryu

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Research Profile

- Incoming M.S. student in Computer Science at NYU Courant (Fall 2026), with undergraduate training at Sejong University and research internship experience in robotics / computer vision.
- Interested in robot learning, manipulation, imitation learning, calibration, perception, real-world deployment, and human-in-the-loop correction.

Education

M.S.	New York University, Courant Institute of Mathematical Sciences , Computer Science	New York, NY
	• Incoming student, Fall 2026.	
Undergraduate Study	Sejong University , Computer Science	Seoul, South Korea
	• Background in computer science, robotics, and computer vision.	

Experience

Robotics and Computer Vision Lab, Sejong University , Research Intern	Seoul, South Korea Jan 2026 – Apr 2026 4 months
• Worked on real-robot manipulation and robot learning pipelines with Piper / SO-101 arms, camera calibration, teleoperation, dataset collection, policy training, and deployment debugging. • Adapted robot learning and calibration tooling to an AgileX Piper single-arm setup, including Piper backend support, leader-follower teleoperation loops, and calibration CLI options. • Set up and debugged real robot systems on Jetson hardware, including CAN communication, Piper SDK issues, SO-101 leader/follower arms, RealSense cameras, and Hugging Face data/model workflows. • Collected teleoperation demonstrations for manipulation tasks, trained smolVLA / Diffusion Policy style models, deployed checkpoints on hardware, and inspected rollout failures. • Built and debugged multi-camera calibration workflows using wrist and third-view cameras, ChArUco boards, hand-eye calibration, extrinsic transforms, and Open3D visualization. • Analyzed deployment robustness under camera viewpoint changes, wrist-camera occlusion, OOD object placement, action-chunk latency, grasp instability, and workspace constraints.	

Projects

End-to-End Manipulation Pipeline with LeRobot and smolVLA

- Collected teleoperation datasets with SO-101 arms, uploaded data to Hugging Face, fine-tuned a pretrained smolVLA model, and evaluated learned behavior under camera and object-placement changes.
- Compared fixed-view and moving-view datasets for a wet-tissue stacking task.
- Defined a subtask-level evaluation rubric for approach, grasp, lift, move, and place.
- Observed strong dependence on wrist-camera input and failures under randomized object placement.

Diffusion Policy Deployment and EE-Space Dataset Conversion

- Set up a real-robot block pick-and-place experiment, collected 100 demonstrations, started Diffusion Policy training, deployed a 40K checkpoint, and converted joint-space trajectories toward end-effector-space representations.
- Investigated action-chunk latency and discontinuous deployment behavior.
- Used forward kinematics to convert trajectory data and identified rotation-vector discontinuities.
- Considered quaternion orientation as a more stable representation for learning.

Piper Calibration and Cross-View Alignment

- Adapted RAIDEN calibration tools to a Piper arm, collected calibration poses, ran ChArUco-based calibration, and visualized camera frames and RGB-D point clouds in Open3D.
- Added Piper-specific backend logic where the original toolkit assumed a YAM bimanual robot.
- Debugged board detection, marker dictionary mismatch, marker sizing, RealSense setup, and transform-chain interpretation.
- Verified scene-camera and wrist-camera placement using base-frame, FK, and hand-eye transforms.

Human-in-the-Loop Joint-Delta Intervention

- Modified a Piper Diffusion Policy intervention path so a human operator could correct policy execution through joint-space delta teleoperation.
- Replaced less accurate delta end-effector intervention with a joint-delta control path.
- Reduced reliance on forced leader/follower synchronization during intervention.
- Preserved the existing policy and recording structure while changing the corrective-control mechanism.

Technical Skills

Robot Learning: imitation learning, diffusion policy pipelines, smolVLA / VLA models, policy deployment, OOD evaluation

Real Robot Systems: Piper arm, SO-101 arm, leader-follower teleoperation, Jetson Orin / Thor, CAN communication, RealSense cameras

Calibration and Perception: hand-eye calibration, camera extrinsics, ChArUco boards, Open3D visualization, multi-view alignment

Engineering Tools: Python, PyTorch, LeRobot, Hugging Face Hub, wandb, Linux / Jetson setup

Research Interests

- Robot learning for manipulation, embodied AI, imitation learning, real-world deployment, calibration, perception, robustness, and human-in-the-loop correction.